

# MAKAR BADERKO

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## ABOUT ME

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Im a 17-year-old software engineer and AI researcher with an interdisciplinary background in mathematics, computer science, economics, and physics. My work focuses on machine learning and data-driven modeling.

## SKILLS

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ML &amp; AI</b>  | PyTorch, Classic ML, Gradient Boosting, Data Mining, Time-series Modelling, Deep Learning, Computer Vision, LLM Fine-Tuning, Graph Neural Networks |
| <b>Quantitative</b> | Mathematical Modelling, Statistical Experiment Design, Stochastic Optimisation                                                                     |
| <b>Economics</b>    | Econometric Analysis, Financial Modelling                                                                                                          |
| <b>Languages</b>    | Russian – Native, English – C1 (Cambridge certified), German – B2                                                                                  |

## RESEARCH

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- AdLoCo: Adaptive batching significantly improves communications efficiency and convergence for Large Language Models - *ICOMP 2025*

## ACADEMIC ACHIEVEMENTS

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### Computer Science and AI

- National Statistics and Data Analysis Olympiad - Gold(2022, 2024, 2025), Silver(2023)  
Top-1 result in the theoretical section (2022, 2024) & all-time national cumulative score record holder  
*This is the most significant olympiad in data analysis in Russia*
- Kaggle Competitions Master – world's top 300 Data Scientist (1x Gold, 2x Silver)
- Autumn National Olympiad in AI (IAIO 2026 National Qualifying Round) – 5th in the country
- National Technological Olympiad in Artificial Intelligence – Gold(2025)  
Top-1 result (among my grade participants)  
*This was the most significant olympiad in AI in Russia at the time of participation*

Eligible for merit-based admission to Russia's top-tier universities (MIPT, HSE, MSU, ITMO)

## WORK EXPERIENCE

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### Laboratory of Mathematical Methods of Optimisation, MIPT

Machine Learning Research Intern

(June 2025 - August 2025)

- Co-developed AdLoCo, a novel distributed optimisation framework that significantly improves communication efficiency and convergence for LLMs
- Contributed to the mathematical proof of communication complexity bounds for non-convex stochastic optimisation

### Educational Scientific Center Sirius

Machine Learning Research Intern

(July 2024 - October 2024)

- Applied Hydra bidirectional and Mamba/Mamba2 architectures to the problem of sequential recommendation, outperforming SOTA models in latency & NDCG
- Managed model fine-tuning on a GPU cluster, utilizing complex custom-developed schedulers and adaptive batching techniques

Granted direct admission to internships in ML at largest russian banks (Sber and Tbank) for winning top corporate ML competitions (interview waived).

## PROJECTS

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### Computer Science and AI

- Machine learning model that uses publicly available blockchain network data to prevent a transaction with a high risk of money laundering (GNNs and more)
- Parcel weight and size estimation by photo and description using large language models and computer vision techniques for Avito, the largest classified ads platform in Russia
- NCAA Basketball matches outcome estimation using past match performance, bookmaker odds and machine learning (Kaggle Gold Medal)

### Economics and Finance

- Governmental investment effectiveness evaluation and prediction based on overall economy performance using data about 200,000 companies
- Private investment regularity evaluation and prediction for brokerage recommender systems using data about 5000 clients